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Plug-and-play method for segmenting concrete bridge cracks using the segment anything model with a fractal dimension matrix prompt

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ABSTRACT

This paper addresses the diverse scenarios of bridge crack segmentation, proposing a method for detecting cracks on land and underwater using the Segment Anything Model (SAM) prompted by a fractal dimension matrix. The proposed method does not require additional training and obtains fractal feature information of cracks through fractal dimension matrix calculation. These feature information serve as prompt information for SAM to establish a plug-and-play crack segmentation method. The method achieves high detection performance, with a mean accuracy, IoU, and F1-Score of 99.6 %, 0.89, and 0.95 for land cracks, and 97.6 %, 0.89, and 0.95 for underwater cracks, respectively. This represents a significant improvement over methods that do not use the fractal dimension matrix for SAM prompts. Additionally, the method requires no additional training, showcasing excellent generalizability and practical potential for real-world applications in diverse environments.

1. Introduction

Health monitoring and maintenance of bridge have become increasingly important in modern society, especially regarding the detection of cracks in concrete structures [1]. Concrete cracks both affect the aesthetics of bridges and pose a serious threat to their structural integrity and longevity [12]. Traditional crack detection methods primarily rely on manual inspection, which is not only inefficient but also susceptible to human error, making it difficult to ensure the accuracy and reliability of the detection results [21,26]. With technological advancements, automated detection methods based on image processing have been gradually introduced. However, these methods often require complex image preprocessing and feature extraction steps, and their performance in practical applications still needs improvement [38].

The deep learning methods have made significant advances in the field of image processing, particularly with convolutional neural networks (CNNs) demonstrating robust capabilities in object detection and image segmentation [24,31]. Deep learning models trained on large amounts of labeled data can achieve high-precision crack detection [19,33,39]. Li et al. [16] proposed a lightweight CNN-transformer hybrid network for semantic segmentation of dam cracks, introducing the transfer learning and depthwise separable convolutions to optimize the model, achieving a segmentation accuracy of 97.60 %. Liu et al. [22] improved tunnel crack detection through an enhanced object detection algorithm combined with image enhancement techniques, resulting in precise crack localization. Wan et al. [35] utilized the Single Shot MultiBox Detector (SSD) algorithm to achieve effective crack detection with high precision and recall rates. To further improve the detection of crack details, some researchers have incorporated additional geometric features of crack images to obtain more comprehensive crack characteristics, thereby enhancing detection accuracy. Liu et al. [19] achieved high-precision detection across multiple crack datasets by integrating 2D spatial features and 3D geometric features. He et al. [11] proposed a boundary-guided crack segmentation model (BGCrack), which outperformed other algorithms in terms of both detection efficiency and performance. Yang et al. [36] proposed an effective concrete crack segmentation network based on UAV-enabled edge computing, the network used feature map fusion to fuse different levels of feature map information into lower-level features for crack segmentation. Based on the statistical analysis results of crack images, Chen et al. [4] constructed a multi-stage feature extraction network with multi-resolution parallel transmission to extract crack features and achieve high-precision segmentation of cracks.

The SAM is a deep learning model pre-trained on extensive datasets

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[13], capable of performing a wide range of segmentation tasks without requiring additional training samples. This model boasts robust generalization capabilities and is considered a universal model [6,18]. Chen et al. [5] utilized the SAM for the segmentation of medical ultrasound images, demonstrating its effectiveness on images with clear tissue structures. Carraro et al. [3] applied the SAM in the agricultural field, validating its potential value across multiple datasets. Yang et al. [37] employed the SAM on various public datasets, achieving results comparable to state-of-the-art fully supervised edge detectors through adaptive progressive learning and gradient-guided pseudo-label updates. Other application cases further support SAM's versatility and generalization ability across different domains [7,14,30]. These examples confirm SAM's capability to address segmentation problems in various fields through zero-shot training. However, they also highlight that SAM's performance is limited in the presence of shadow artifacts and blurred boundaries. The key to overcoming this limitation lies in providing SAM with more precise prompt information.

The fractal geometry effectively describes the complex shapes and structural characteristics found in nature, and fractal dimensions can quantify image complexity, provides a high-quality choice for prompt information of SAM. The fractal theory has been widely applied across various fields. Duan et al. [8] summarized the contributions of fractal theory to the study of packaging material properties. Guan et al. [10] proposed a surface roughness measurement method based on microscopic images and fractal theory, where calculating fractal dimensions effectively represented surface conditions. In the field of image analysis, Ristanović et al. [28] used fractal dimensions to characterize dendritic morphologies. Panigrahy et al. [25] analyzed image textures using fractal dimensions, demonstrating its effectiveness across multiple datasets. Crack shapes are often highly variable, and fractal theory can compute image complexity to describe crack features, providing a novel approach to obtaining crack characteristic information. Su et al. [32] calculated the dimensional features of cracks using the fractal theory and embedded these features into a semantic segmentation model, validating the approach across multiple building structure crack datasets.

In summary, the current deep learning-based crack segmentation methods still face certain limitations in practical applications. Training deep learning models requires a large amount of labeled data and high computational costs. Additionally, these models have limited generalization capabilities, making it difficult to adapt to diverse application scenarios, especially when detecting cracks in different environments where performance can vary significantly. The application of SAM can effectively address these issues, but achieving optimal detection accuracy requires effective physical information to prompt SAM. Fractal theory, with its powerful shape feature description capabilities, can provide more robust prompts for the SAM. Therefore, this paper proposes a crack segmentation method based on a fractal dimension matrix to prompt the SAM for detecting concrete cracks both on land and underwater. The main contributions are as follows:

(1) Improved Box-Counting Method: The fractal dimension matrix obtained through this method can calculate fractal dimensions across multiple scales and spatial regions, offering a more comprehensive, detailed, and robust analysis of complex structures. This significantly enhances the accuracy and efficiency of crack detection.
(2) Plug-and-play Model for Crack Image Detection: The proposed method does not require additional training processes, making it theoretically applicable to any crack images, thus providing a universal model for crack image detection.

The organization of the paper is as follows: The Section 1 (Introduction) conducted research on domestic and foreign studies, indicating the research objectives for this study; Section 2 (Method) provides a detailed explanation of methods such as data acquisition, fractal dimension matrix calculation, crack segmentation models, and image enhancement; Section 3 (Results and Discussion) presented, analyzed, and discussed the segmentation results; Section 4 (Conclusion), the research results were organized to obtain guiding research conclusions, and future research directions were pointed out.

2. Method

This paper calculates the fractal dimension matrix of images to obtain feature information related to cracks, which is then used as prompt information for SAM in crack segmentation. Simultaneously, to achieve the detection of underwater cracks in concrete bridges, this paper employs a crack image enhancement method to improve the quality of underwater crack images, thereby achieving more accurate crack detection results. In summary, this paper proposes a plug-and-play crack segmentation method for detecting both land and underwater cracks (Fig. 1).

2.1. Underwater crack image acquisition

The underwater robot (BG-420) developed in this paper was used to collect underwater crack images [20], as shown in Fig. 2a, which consists of propellers, battery, camera, control cabin, remote control, and cable. Its body size was 807 mm × 500 mm × 127 mm, using the high-strength aluminium alloy material with characteristics of corrosion and impact resistance. The BG-420 underwater robot has a maximum diving depth of 200 m, uses 8 thrusters (which can provide vertical and horizontal propulsion), a speed of 2.5–3.5 m/s, the PID dual loop fuzzy control algorithm [17] was used to implement motion control, and can resist water flow up to 2 m/s, an 12-megapixel camera with 2 × 15 W LED lights mounted on the BG-420 captures detailed images of underwater structures, essential for visual inspections and enabling the detection and documentation of defects such as cracks, spalling, and rebar exposure. The high-resolution camera ensures clear underwater images and videos for real-time monitoring and post-mission analysis. It also uses DDPG-based [40] disturbance flow correction control algorithm to solve the impact of turbulence disturbance, greatly reducing the threshold for manual operation, improving the intelligence level of equipment, and adapting to various complex water flow environments. The image acquisition path and process were shown in Fig. 2b, the collection path was executed through the implementation of path planning algorithms, using a loop up and down path to complete pier collection.

The underwater robot PID dual-loop fuzzy control algorithm used in this paper combines the advantages of traditional PID control and fuzzy control to improve the control accuracy and response speed of the underwater robot. The dual-loop control structure includes inner and outer control loops. The inner loop uses PID control to achieve fast response and system stability. The basic equation was:

$$u(t) = K_p e(t) + K_i \int e(t)dt + K_d \frac{de(t)}{dt} \quad (1)$$

The outer loop uses a fuzzy controller to adjust the PID controller's

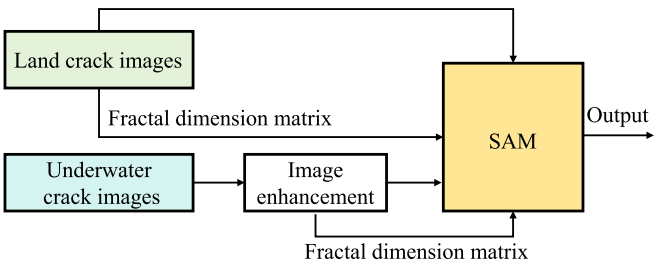


Fig. 1. Research outline.

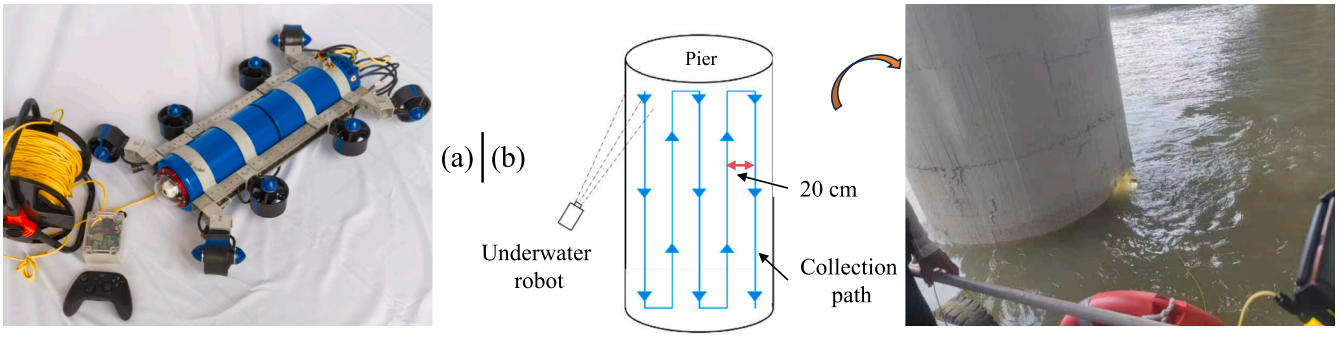


Fig. 2. Underwater robot (BG-420) and image acquisition process.

parameters. The fuzzy controller, based on the error e and the rate of change of the error Δe, outputs the PID parameter adjustments ΔK_p, ΔK_i and ΔK_d through the processes of fuzzification, fuzzy inference, and defuzzification. The adjusted PID parameters are:

$$K_p = K_p0 + \Delta K_p \tag{1}$$

$$K_i = K_i0 + \Delta K_i \tag{2}$$

$$K_d = K_d0 + \Delta K_d \tag{2}$$

Where, K_p0, K_i0, and K_d0 are initial parameters of PID.

2.2. Fractal theory and fractal dimension matrix

The fractal theory was a mathematical discipline that studies irregular and complex geometric shapes, introduced by the French mathematician Benoît Mandelbrot in the 1970s. The fractal theory has widespread applications across various fields. In natural phenomenon simulation, fractals were used to describe and analyze complex structures such as coastlines, mountains, clouds, and trees. In computer graphics, fractal algorithms generate realistic natural images and textures, including terrains, plants, and clouds. In signal and image processing, fractal theory was applied in image compression, texture analysis, and edge detection [29]. The fractal theory provides a powerful tool for understanding and describing the natural world and various complex systems by revealing self-similar properties and multi-scale structures within these systems. The core concepts of fractal geometry include self-similarity, iterative processes, and fractal dimension [27]. Self-similarity refers to the similarity of fractal structures at different scales, which can be either strict or statistical. The iterative process involves generating complex shapes by repeatedly applying simple rules or formulas [9]. The fractal dimension quantitatively describes the complexity of fractals and was typically a non-integer dimension. The box counting dimension was one of the most commonly used fractal dimensions. It calculates the dimension value by covering the space of the fractal object with grids (boxes) of different sizes, as shown in Eq. 1 and Fig. 3. Overall, fractal theory, by elucidating the self-similar nature and multi-scale structures of complex systems, offers a robust framework for modeling and analyzing a wide range of phenomena, providing insights into both natural and artificial forms.

$$D = \lim_{\epsilon \to \infty} \frac{log(N(\epsilon))}{log(1/\epsilon)} \tag{1}$$

Where, $N(\epsilon)$ represents the number of boxes required to cover the fractal, ϵ is the edge length of each box.

The standard box counting method for calculating fractal dimensions relies on binary images [23], which can lead to the omission of many details during the calculation process. For images with complex

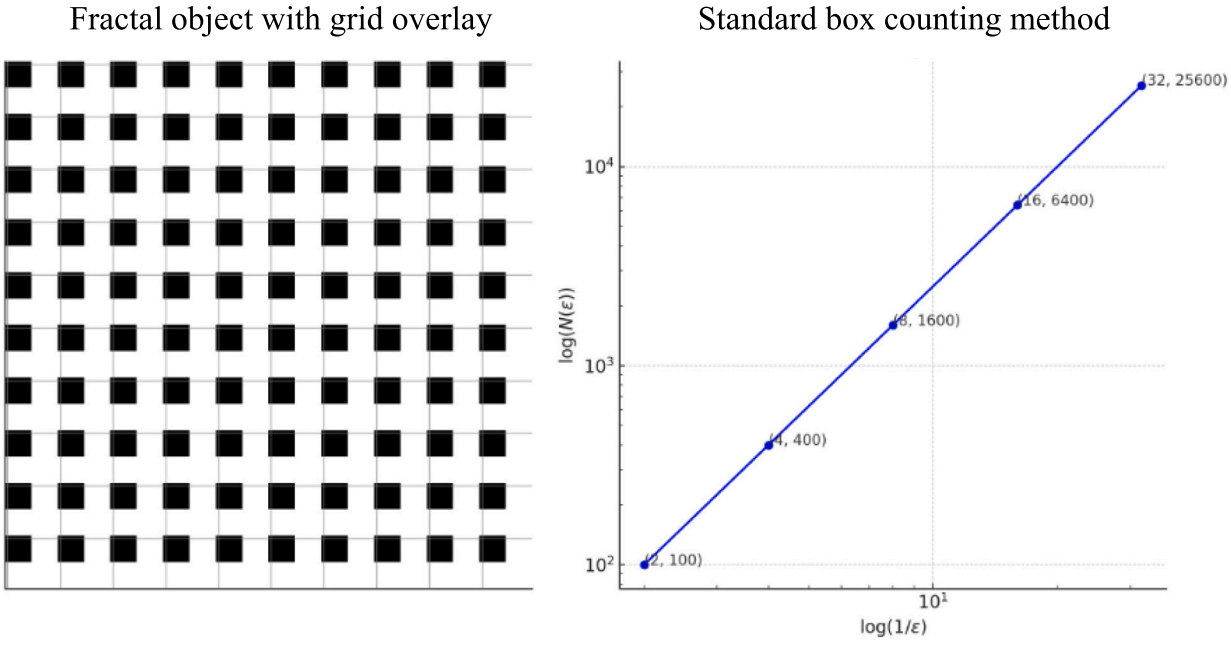


Fig. 3. Standard box counting method.

structures or multi-scale features, the standard box counting method may fail to accurately capture their fractal characteristics. Additionally, the noise can cause more boxes to be counted as non-empty, thereby affecting the final fractal dimension calculation and potentially leading to an overestimation of the fractal dimension. Crack images often contain complex information and may include significant noise. Therefore, the standard box counting method was not suitable for crack images. To improve the accuracy of fractal dimension calculation for crack images, this paper proposes an improved box counting method. The calculation method was as follows (Eq. 2):

$$D = -\lim_{T_i \to \infty} \frac{log(N(T_i))}{log(T_i)} \tag{2}$$

Where, T_i represents a specific threshold within the grayscale threshold set(For each grayscale threshold T_i, the image undergoes binarization, where pixels in the original grayscale image greater than or equal to T_i are set to 1, and those less than T_i are set to 0) , $N(T_i)$ denotes the number of non − empty boxes in the binary image obtained after applying the grayscale threshold T_i.

The improved method was applicable for processing grayscale images, not just binary images. Grayscale images retain more detailed information, enabling more accurate calculation of fractal dimensions [15], the box counting method for grayscale images was shown in Fig. 4. By applying multiple grayscale thresholds to the image, the improved method can capture crack details across different grayscale levels. This enables a more comprehensive analysis of the complexity and structural features of cracks. The use of multiple thresholds enhances the robustness of the calculation process against noise and local variations, allowing for a more stable extraction of the overall structural characteristics of cracks. As different grayscale thresholds reveal various levels of detail within crack images, the improved method can extract crack edges and internal structures more meticulously.

However, a single fractal dimension can only provide an average measure of the overall complexity of a structure, making it difficult to reflect variations in detail and features across different scales. Fractal dimension matrices serve as multi-scale, multi-directional tools for fractal analysis, aiming to describe the internal structure and features of complex systems. By computing local fractal dimensions in the spatial domain, it becomes possible to reveal variations in complexity and fractal characteristics across different regions of an image, providing spatial distribution information. Additionally, multiple local fractal dimensions can reflect changes in detail and texture features across the image, offering a detailed depiction of complexity in different regions. In summary, fractal dimension matrices enhance complex structural analysis capabilities by computing fractal dimensions across multiple scales and spatial regions. Compared to single fractal dimensions, they can more accurately capture multi-scale features, spatial distribution information, and variations in detail, presenting a significant advantage in fractal feature extraction from crack images. The calculation of local fractal dimensions was represented by Eq. 3.

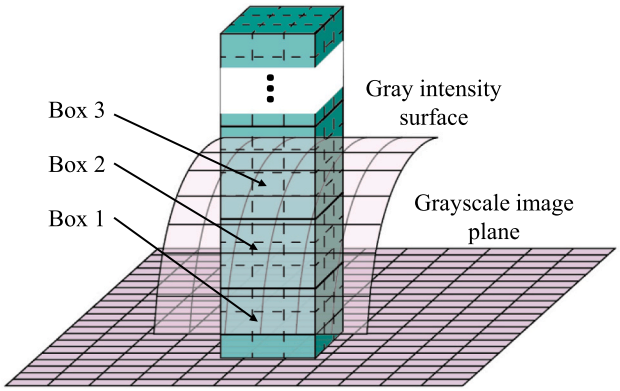


Fig. 4. Box counting method for grayscale images.

$$D(q,s) = -\lim_{T_i \to \infty} \frac{ln(N_q(T_i,s))}{ln(T_i)} \tag{3}$$

Where, $N_q(T_i,s)$ represents the number of boxes covering the fractal object at scale s (the sizes of different colored blocks in Fig. 5), where the number of objects in each box is weighted according to the parameter q. The parameter q is introduced to control the weighting of box counting for objects, thus revealing the fractal structure across different scales. By arranging the local fractal dimensions, a fractal dimension matrix of the image can be obtained (Fig. 5, different colors represent different regions).

The algorithmic steps for fractal dimension matrix:

Input: Grayscale image *I_gray* of size [*rows*, *cols*]. Block size *blockSize* to divide the image into smaller sub-blocks.

Output: *fractalDimMatrix*: A matrix containing the calculated fractal dimension for each sub-block of the image.

Step 1: Initialize Parameters.

1.1. Get the size of the grayscale image *I_gray* as [*rows*, *cols*].

1.2. Initialize *fractalDimMatrix* with zeros, where the matrix size is [floor(*rows/blockSize*), floor(*cols/blockSize*)].

Step 2: Divide Image into Blocks.

For each block starting at position (*i*, *j*) in *I_gray*, where *i* and *j* are incremented by *blockSize*:

Extract a sub-block of size *blockSize* × *blockSize* from *I_gray*.

Step 3: Calculate Fractal Dimension for Each Block.

For each extracted sub-block, use the *improvedBoxCount* method to calculate its fractal dimension:

Convert the sub-block to a double-precision matrix.

Determine the maximum and minimum gray-level values in the block.

Define a range of thresholds between the minimum and maximum gray levels.

Initialize *countBoxes*, which records the number of "boxes" (non-zero elements) for each threshold.

For each threshold, create a binary image by applying the threshold to the block and count the number of non-zero elements (boxes).

Step 4: Fit the Log-Log Data.

4.1. After computing *countBoxes* for all thresholds, remove the zero values from *countBoxes* and their corresponding threshold values.

4.2. Perform a linear regression on the log-transformed values of countBoxes and threshold.

4.3. The slope of the fitted line from the log-log plot gives the fractal dimension *D*.

Step 5: Store Fractal Dimension.

Store the computed fractal dimension *D* in the corresponding position in *fractalDimMatrix*.

Step 6: Return Result.

Once all blocks are processed, return the matrix *fractalDimMatrix* containing the fractal dimensions of all sub-blocks.

2.3. Segment anything model

The image segmentation was a fundamental task in computer vision, crucial for applications ranging from medical imaging to autonomous driving. Traditional segmentation models typically require training for specific tasks and were limited in their applicability across different domains. The SAM addresses these limitations by offering a universal solution adaptable to many segmentation tasks without the need for retraining. Developed by Meta AI, the SAM was a novel image segmentation framework with high flexibility and generalizability. The SAM leverages various input cues such as points, bounding boxes, and textual descriptions, enabling users to perform segmentation tasks accurately and effortlessly across diverse applications. The architecture

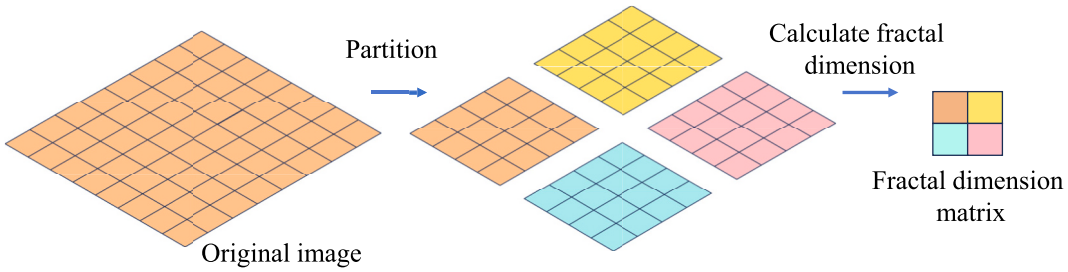


Fig. 5. Calculation process of fractal dimension matrix.

of SAM primarily consists of three components (Fig. 6): the image encoder (Vision Transformer (ViT)), prompt encoder, and mask decoder.

The SAM utilizes the ViT as the image encoder (Fig. 7), serving as the backbone network for image feature extraction. The choice of ViT architecture stems from its ability to capture long-range dependencies and intricate visual information. Key components of the ViT backbone include:

(1) Patch Embedding: Dividing the input image into fixed-size patches (Eq. 4), each patch was flattened and linearly embedded into a high-dimensional space.

The input image $x \in \mathbb{R}^{H\times W\times C}$ was divided into a sequence of N non-overlapping patches. Each patch was of size $P \times P \times C$, where H and W are the height and width of the image, and C is the number of channels. The number of patches N is given by:

$$N = \frac{H \times W}{P^2} \tag{4}$$

(2) Transformer Encoder: A series of Transformer encoder layers process the embedded patches. These layers consist of multi-head self-attention mechanisms and feed-forward neural networks, enabling the model to learn complex features from the input image.

The sequence of embedded patches is then fed into a Transformer encoder. The Transformer encoder consists of multiple layers, each containing Multi-Head Self-Attention (MSA) and a Feed-Forward Neural Network (FFN). For the $l-th$ layer, the operations are as follows:

$$z_l = MSA(LN(z_{l-1})) + z_{l-1} \tag{5}$$

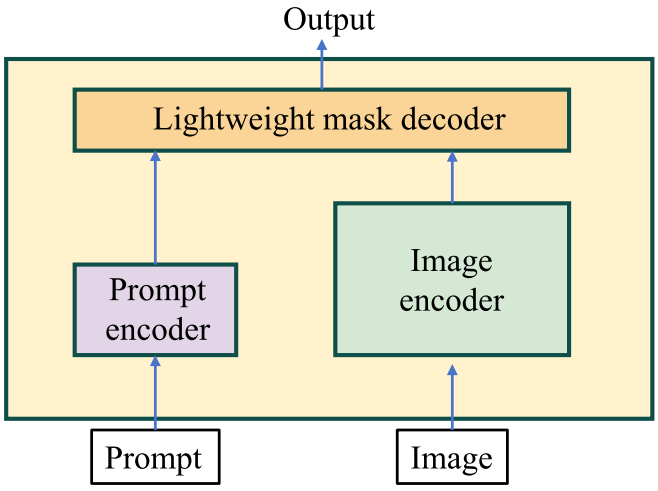


Fig. 6. Composition of SAM.

$$z'_l = MSA(LN(z_l)) + z_l \tag{6}$$

Where, LN denotes Layer Normalization, and z_l represents the output at layer l.

(3) Position Embedding: Adding positional information to the patch embeddings to preserve spatial relationships within the image. These components collectively empower the ViT to effectively extract rich visual features from input images, making it well-suited for various tasks including image segmentation in the SAM.

Each patch is then flattened into a vector and projected to a fixed dimension D using a linear projection. This can be represented as:

$$z_0^i = Linear\left(x^i\right) + E_{pos}^i, i = 1, \ldots\ldots, N \tag{7}$$

Where, E_{pos}^i is the positional embedding, which encodes the position of each patch.

The prompt encoder processes user inputs (points, boxes, text, and masks) to guide the segmentation process. Specifically, it includes: (1) Point Encoding: User-provided points were encoded into vectors representing the position and type (positive or negative sample) of the points. (2) Box Encoding: User-drawn bounding boxes were encoded to provide spatial constraints for segmentation. (3) Text Encoding: User-provided textual descriptions were transformed into feature vectors using pre-trained language models. These vectors assist the model in understanding the context of the segmentation task. (4) Mask Encoding: User-provided initial segmentation masks were encoded to refine or improve existing segmentation results. Mask prompts were embedded using convolutions and element-wise summed with image embeddings.

The mask decoder integrates features from both the ViT backbone and the prompt encoder to generate segmentation masks. The decoder comprises: (1) Feature Fusion: Utilizing attention mechanisms to combine features from the ViT backbone and the prompt encoder, ensuring that segmentation was accurately guided by user input. (2) Upsampling Layers: Gradually upsampling the fused features to the original image resolution, generating high-resolution segmentation masks. (3) Mask Generation: Applying a sigmoid function to the output of the final upsampling layer to generate binary masks, defining the segmented regions.

The pre-training process of SAM occurs on a large-scale and diverse dataset (SA-1B dataset), which comprises over a billion finely annotated images and segmentation masks covering various image types and segmentation tasks. The pretraining process, illustrated in Fig. 8, adopts a multitask learning strategy, combining cross-entropy loss and Dice loss to optimize segmentation accuracy. Data augmentation techniques such as random cropping, scaling, and rotation were employed to enhance the model's robustness to different image transformations. The objective of pretraining was to enable SAM to learn rich feature representations, allowing it to generalize to different segmentation tasks and application scenarios without the need for retraining.

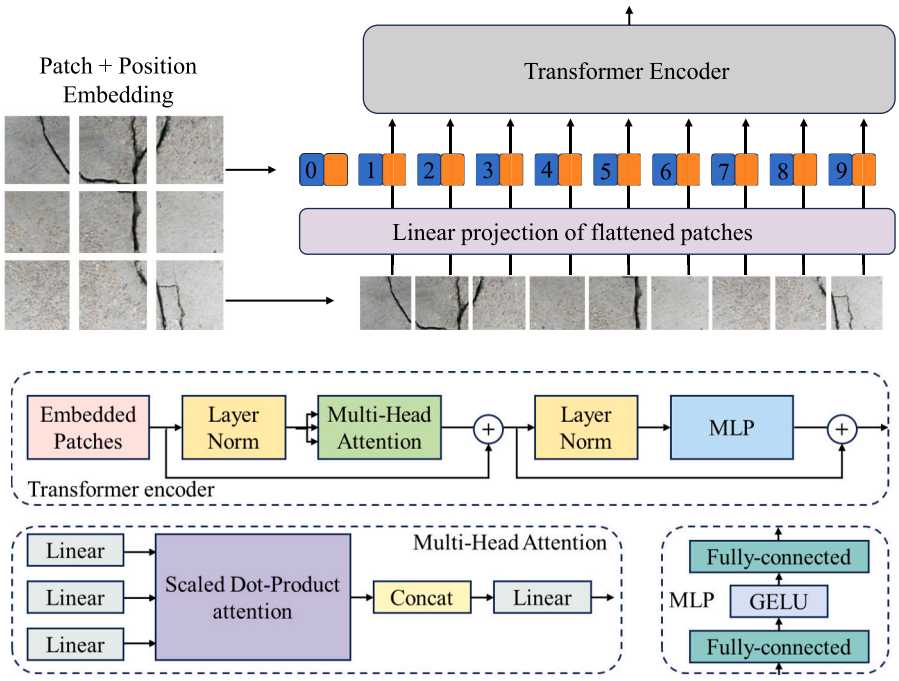


Fig. 7. ViT backbone.

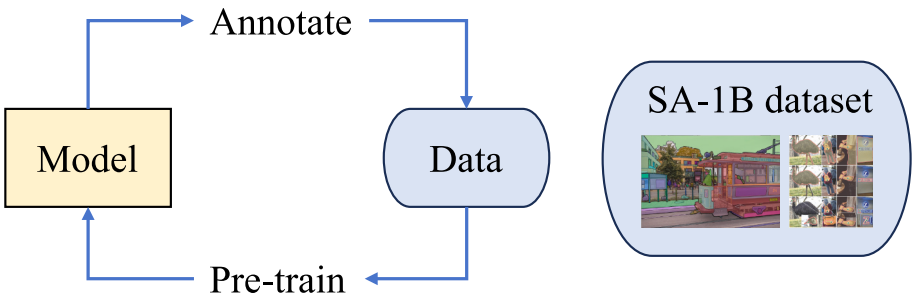


Fig. 8. Pre-training process of SAM.

2.4. Crack detection method

Crack detection was a crucial task in infrastructure maintenance. As the SAM was a universal segmentation model, it eliminates the need to retrain models for each material or type of crack, greatly simplifying the deployment process. In terms of network architecture, the ViT backbone of SAM can handle high-resolution images, capturing subtle features of cracks. Additionally, the SAM allows for the embedding of prompt information, which can guide the SAM to achieve more accurate detection results. Therefore, if preliminary crack feature information was available, users can provide initial prompt information, and the SAM can further refine and improve these results. As described in Section 2.1, the fractal dimension matrix encompasses multi-scale features, spatial distribution information, and detail variations of crack images, making them reliable crack feature prompt information. Hence, this paper proposes using fractal dimension matrix as SAM's prompt mask to

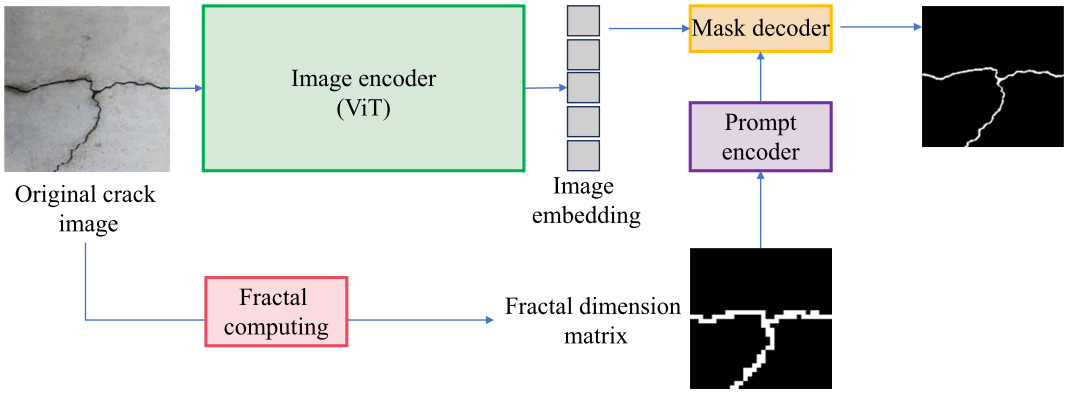


Fig. 9. Proposed crack detection method.

provide a approach for crack segmentation, as illustrated in Fig. 9, depicting the implementation process of the proposed method.

The effectiveness of the proposed method will be validated through crack images from two significantly different scenarios: on land and underwater. Since the proposed method does not require training samples, this study has prepared 165 land crack images and 15 underwater crack images (collecting them through underwater robots) as the segmentation targets. The land crack images are of good quality, while the underwater crack images suffer from low brightness and contrast due to lighting conditions, turbidity, and contain a significant amount of noise. Therefore, image enhancement operations will be applied specifically to the underwater crack images.

2.5. Underwater image enhancement

In underwater environments, images captured typically exhibit low contrast and brightness due to light scattering and absorption [34], making it challenging to discern details such as cracks [2]. To enhance the quality of underwater crack images, this paper adopts an enhancement method based on image inversion and haze removal. Firstly, the input low-light underwater crack image undergoes an inversion operation (Eq. 8), transforming it into a form similar to a hazy image. Subsequently, haze removal processing was applied to the inverted image, enhancing the contrast and details by estimating and removing the scattered light component from the image (the haze image model established as Eq. 9, combined with the calculation value of $t(x)$ (Eq. 10), and the known $R(x)$, A, the haze-free image $J(x)$ was calculated as Eq. 11). Finally, the image after haze removal undergoes inversion again (Eq. 12) to restore the image with normal brightness and contrast (I_{output}). Through this dual inversion and haze removal process, the details and contrast of underwater crack images were significantly enhanced, enabling more accurate and reliable detection and analysis of cracks. This method was particularly suitable for low-light underwater environments with scattering effects, effectively improving image quality and assisting in maintenance and monitoring tasks of underwater structures.

$$R^c(x) = 255 - I^c(x) \tag{8}$$

Where, $R^c(x)$ represents the pixel value at position x in the image after inversion, and $I^c(x)$ represents the pixel value at position x in the original low-light image. c denotes the color channel: red, green, or blue.

$$R(x) = J(x)t(x) + A(1 - t(x)) \tag{9}$$

Where, $R(x)$ denotes the intensity of pixel x captured by the camera. $J(x)$represents the intensity of the original object in the scene. $t(x)$ is the transmission map, describing the proportion of light emitted from the scene that is retained when reaching the camera. A stands for the global atmospheric light.

$$t(x) = 1 - \omega \min_{c\in(r,g,b)} \left(\min_{y\in\Omega(x)} \left(\frac{R^c(y)}{A^c} \right) \right) \tag{10}$$

Where, ω is the weight coefficient, typically set to. $\Omega(x)$ represents the local region centered at x. $R^c(y)$:denotes the pixel value at position y in the reversed image. A^c represents the atmospheric light in the color channel.

$$J(x) = \frac{R(x) - A}{t(x)} + A \tag{11}$$

$$I_{output} = 255 - J(x) \tag{12}$$

2.6. Evaluation indicator

In this paper, three performance indicators were used to evaluate the detection performance of the semantic segmentation model, including: (a) Accuracy, which calculates the accuracy of crack segmentation, Eq. 13; (b) F1-Score, a comprehensive metric that combines Precision and Recall, Eq. 14; (c) Intersection over Union (IoU), which assesses the overlap ratio between predicted results (A_p) and ground truth labels (A_r) pixels, Eq. 15.

$$Accuracy = \frac{1}{2}\left(\frac{TP}{TP + FP} + \frac{TN}{FN + TN} \right) \tag{13}$$

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{14}$$

$$IoU = \frac{area(A_p \cap A_r)}{area(A_p \cup A_r)} \tag{15}$$

TP (True Positives): represents the number of crack pixels correctly predicted as crack pixels by the model. *TN* (True Negatives): represents the number of non-crack pixels correctly predicted as non-crack pixels by the model. *FP* (False Positives): represents the number of non-crack pixels incorrectly predicted as crack pixels by the model. *FN* (False Negatives): represents the number of crack pixels incorrectly predicted as non-crack pixels by the model. $Precision = TP/(TP + FP)$; $Recall = TP/(TP + FN)$.

3. Result and discussion

This section mainly covers the following five aspects: Section 3.1 demonstrates the effect of underwater crack image enhancement through image inversion and haze removal processing. Section 3.2 reveals the calculation results of the fractal dimension matrix of crack images. Section 3.3 validates the effectiveness of the proposed method in detecting onshore cracks. Section 3.4 validates the effectiveness of the proposed method in detecting underwater cracks. Section 3.5 compares the proposed method with other mainstream deep learning methods.

3.1. Underwater image enhancement results

Fig. 10 illustrates the effects of different image enhancement methods on underwater crack images, particularly showcasing their performance in low-light and scattering environments. The original low-light underwater crack image exhibits low contrast and brightness due to light scattering and absorption, making the crack details difficult to discern. The second row of images displays the results after inversion and dehazing processes. The results indicate that the proposed methods significantly improve the contrast and clarity of the images, making the crack details more pronounced. The histogram stretching enhances image brightness, the adaptive histogram equalization improves image contrast, and the combination of histogram stretching and adaptive histogram equalization enhances both brightness and contrast of the images.

By comparison, it was evident that the proposed enhancement method, based on image inversion and dehazing, excels in improving the brightness and contrast of underwater crack images. The combination of inversion and dehazing significantly enhances image quality, making crack details clearer and facilitating further segmentation and analysis. In contrast, other traditional enhancement methods, while improving image contrast to some extent, fail to achieve the same level of detail enhancement. This may be because traditional methods typically process grayscale images, potentially losing color and detail information during the grayscale conversion, which results in suboptimal enhancement effects.

3.2. Calculation results of fractal dimension matrix

Fig. 11 presents six original images of concrete cracks on land (top half) alongside their corresponding fractal dimension matrix (bottom half). Each image in the top half depicts cracks of varying shapes and complexities. For instance, the crack in the first image has an irregular,

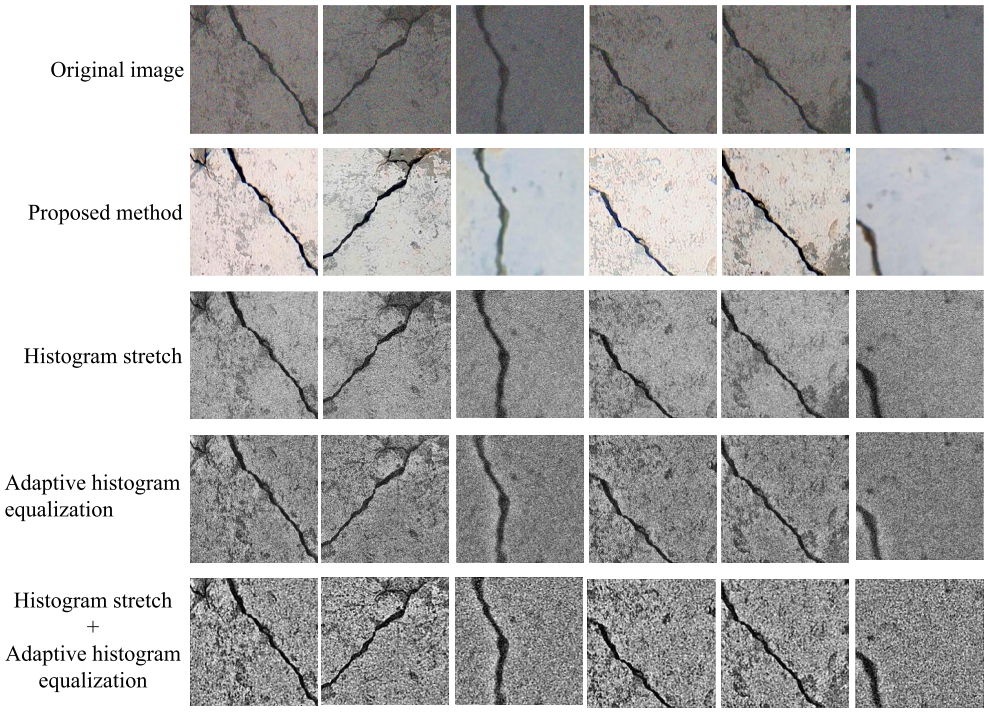


Fig. 10. Enhancement effect of different enhancement methods on underwater crack images.

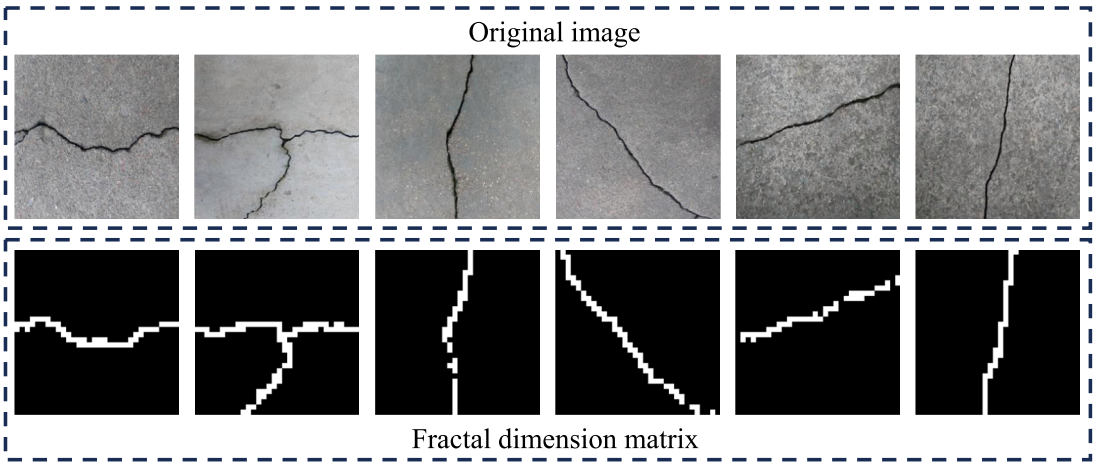


Fig. 11. Calculation results of fractal dimension matrix for cracks in land concrete.

multi-directional path; the second image shows a branched crack; the third image features a relatively straight crack; the fourth image displays a long, curved crack; the fifth image has a slender but not highly twisted crack; and the sixth image shows a mostly straight crack with minor bends. The fractal dimension matrix in the bottom half quantifies the geometric complexity of these cracks, reflecting their details and irregularities. Specifically, the first fractal dimension image shows a high fractal dimension, indicating a complex geometric shape; the second fractal dimension image has a moderate fractal dimension; the third fractal dimension image has a low fractal dimension; the fourth fractal dimension image has a high fractal dimension; the fifth fractal dimension image has a moderately low fractal dimension; and the sixth fractal dimension image has a low fractal dimension. A higher fractal dimension corresponds to greater crack complexity and detail, while a lower fractal dimension indicates simpler crack shapes. These images and their fractal dimension matrices provide better insights into the characteristics of concrete cracks. The fractal dimension results suggest that the fractal dimension matrix can preliminarily describe the shape of cracks. Although it may not accurately capture every detail, it offers useful guidance for the next step in high-precision segmentation processes.

Fig. 12 showcases the changes in underwater crack images before and after enhancement, along with their corresponding fractal dimension matrix. The enhanced crack images (Fig. 12a) exhibit clearer crack edges and details, with the enhancement process making the complex structures and fine branches of the cracks more apparent. The corresponding fractal dimension matrix (Fig. 12b) reflects a high level of geometric complexity, indicating rich crack details. Although it can roughly depict the crack shape, some adverse interference still exists. In contrast, the pre-enhancement crack images (Fig. 12c) have blurred edges and indistinct details, making the complex structures of the cracks hard to discern. The corresponding fractal dimension matrix (Fig. 12d) shows lower geometric complexity, with the crack details and branches being less apparent in the matrix. Compared to the fractal dimension matrix calculated after image enhancement, there are many interference

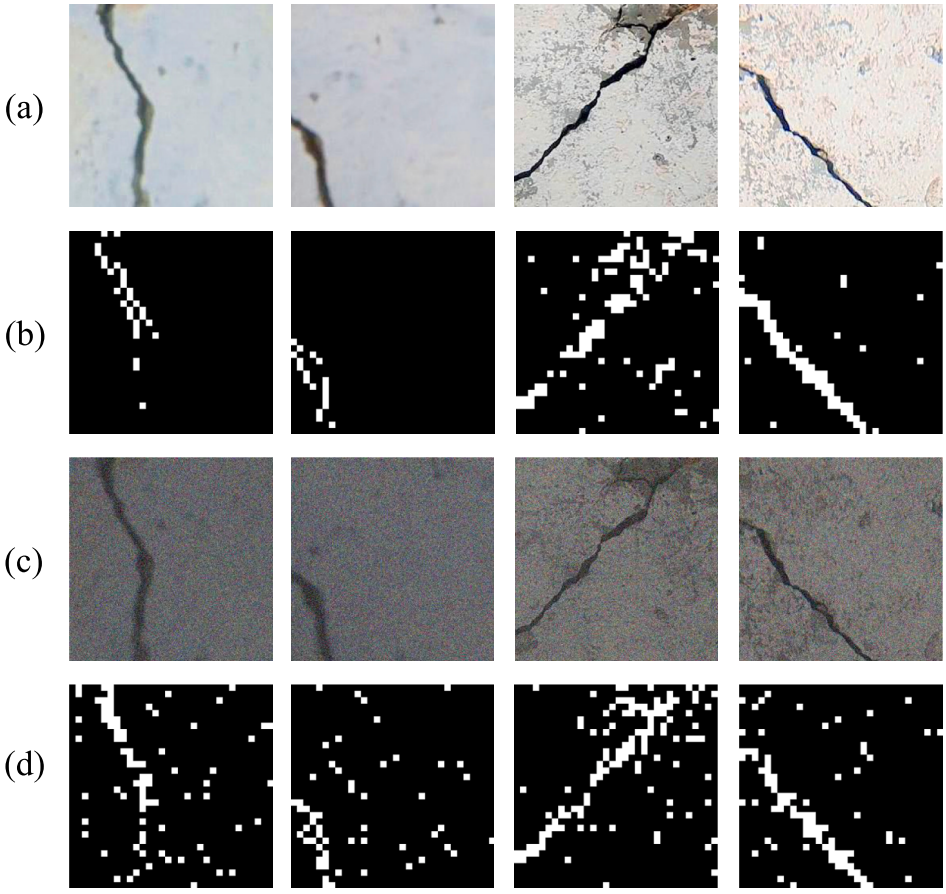


Fig. 12. Fractal dimension matrix of underwater crack images before and after enhancement. (a) Enhanced images, (b) fractal dimension matrix of enhanced images, (c) original images, (d) fractal dimension matrix of original images.

points due to adverse underwater conditions. By comparing the crack images and their fractal dimension matrix before and after enhancement, it is evident that image enhancement significantly improves the clarity and detail of the cracks. The fractal dimension matrix better reflects the geometric complexity and details of the cracks, providing more accurate data support for subsequent crack segmentation processes.

Although the enhancement process improves the accuracy of the fractal dimension matrix in describing crack shapes, there was still some interference, primarily due to contrast, lighting, and the complex texture of the surface. Unlike the fractal dimension matrix results for concrete cracks on land, processing underwater cracks was more challenging due to numerous noise points. Consequently, this presents a significant challenge for subsequent segmentation algorithms.

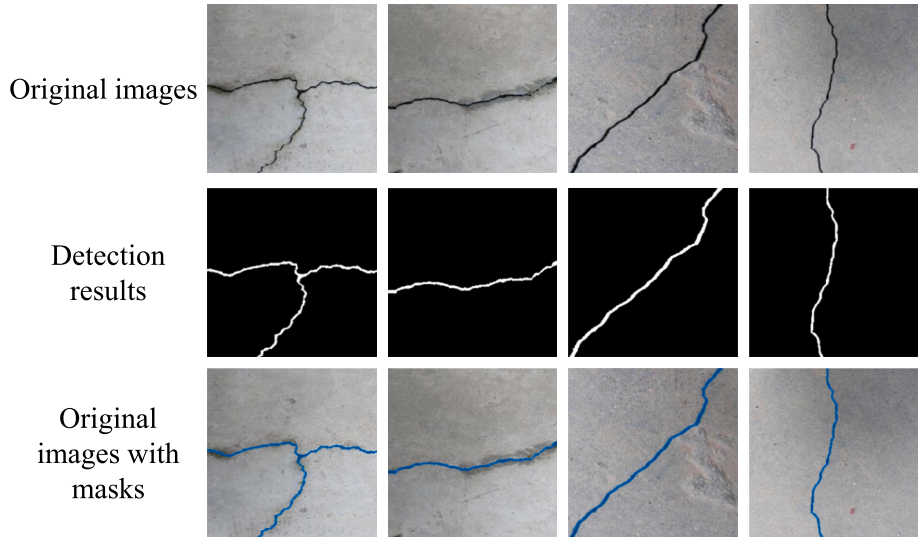


Fig. 13. Examples of segmentation results.

3.3. Results of land crack detection

By using the fractal dimension matrix as a prompt for the SAM, this paper achieved effective segmentation of cracks in land concrete surfaces. Fig. 13 presents examples of the segmentation results for some concrete cracks. It was evident from the images that the SAM performs well in detecting cracks. Notably, the SAM can disregard non-crack dark areas and accurately detect fine cracks, maintaining good consistency with the labels. The detection results for all 165 samples were summarized in Table 1, showing a mean accuracy of 99.6 %, a mean IoU of 0.89, and an F1-Score of 0.95. To highlight the positive impact of incorporating the fractal dimension matrix, this paper also compared these results with those obtained without using the fractal dimension matrix as a prompt. The results without the fractal dimension matrix show a significant decrease in detection accuracy, with a mean accuracy of 84.1 %, a mean IoU of 0.47, and an F1-Score of 0.39. Fig. 14 displays the confusion matrix comparison of detection accuracy (as shown in Fig. 14a, 10,441,345 background pixels were detected correctly with an accuracy rate of 99.7 %, and 32,881 background pixels were detected as crack pixels with a rate of 0.3 %, 124,421 crack pixels were detected correctly with an accuracy rate of 99.7 %, and 689 crack pixels were detected as background pixels with a rate of 0.5 %). From the confusion matrix, it was clear that the detection accuracy improves by 24.1 % when the fractal dimension matrix was used as a prompt. These comparative results confirm that using the fractal dimension matrix to prompt SAM effectively enhances the crack detection performance.

Fig. 15 shows the evaluation indicator for all the detected samples, indicating that the SAM prompted with the fractal dimension matrix was effective for all samples. Compared to methods without fractal dimension information, using the fractal dimension matrix as a prompt achieves higher detection accuracy, IoU, and F1-Score. Fig. 16 illustrates some comparative segmentation cases. The results demonstrate that without the fractal dimension matrix prompt, most detection results contain numerous noise points. This issue arises due to the complex texture of the concrete surface, where non-crack textures were incorrectly detected as cracks, and some actual cracks are missed. Conversely, when using the fractal dimension matrix to prompt the SAM, the detection results are nearly perfect. The method accurately detects crack regions without noise interference and segments details precisely, especially for finer cracks.

3.4. Underwater crack detection results

Using the fractal dimension matrix as a prompt for the SAM in detecting underwater crack images, Fig. 17 showcases the detection results before and after image enhancement. The enhanced underwater images appear clearer, with more prominent details. This enhancement process improves the visibility of cracks, making them more noticeable in the images. The detection results reveal that after enhancement, the effectiveness of crack segmentation significantly increases, with the detected crack edges becoming sharper and more accurate. This indicates that image enhancement aids in improving the performance of the crack segmentation algorithm, allowing for more precise identification and annotation of crack edges and details. In contrast, the original, unenhanced images have lower contrast and fewer details, making the cracks less apparent. Under these conditions, crack segmentation becomes more challenging due to noise and other details in the image that may interfere with the detection results. The detection results from the unenhanced images show less clear crack edges and a higher rate of false detections and missed cracks, especially in darker areas where large patches of incorrect detections occur. The comparative analysis demonstrates that image enhancement plays a crucial role in crack segmentation. Enhanced images not only make cracks more visible but also significantly improve the accuracy and reliability of the segmentation algorithm. Therefore, in practical applications, it was essential to perform appropriate image enhancement to achieve better crack detection results.

Table 1
Comparison of detection results with/without fractal dimension matrix as prompts.

Prompt information	Evaluation indicator		
	Mean accuracy	Mean IoU	F1-Score
Prompt by fractal dimension matrix	99.6 %	0.89	0.95
No-prompt	84.1 %	0.47	0.39

Fig. 18 illustrates the crack detection results using the SAM with and without the fractal dimension matrix prompt. The detection results with the fractal dimension matrix prompt demonstrate a significant improvement, with crack contours accurately marked. The fractal dimension matrix prompt helps the SAM better identify crack edges, resulting in very clear detection outcomes with minimal noise and false detections. This indicates that the fractal dimension matrix prompt significantly enhances crack segmentation, enabling the algorithm to capture crack details more accurately. In contrast, the detection results without the fractal dimension matrix prompt exhibit more noise and false detections. The crack contours in these results are not as clear, with edges sometimes appearing broken or blurry. Additionally, some false crack detections are present, indicating areas incorrectly detected as cracks. This shows that without the fractal dimension matrix prompt, the detection method was more susceptible to interference from complex backgrounds, leading to suboptimal results. The comparative analysis highlights the significant role of the fractal dimension matrix prompt in crack segmentation. The results with the fractal dimension matrix prompt both improve the accuracy and precision of crack segmentation and effectively reduce noise and false detections, making the outcomes more reliable and practical. Therefore, in crack segmentation tasks, applying the fractal dimension matrix prompt was highly valuable, as it can substantially enhance the performance of the segmentation algorithm.

The summary of evaluation indicators for the detected underwater crack image samples was presented in Table 2. The results show that the combination of fractal dimension matrix prompts and image enhancement performs the best, with an average accuracy of 97.6 %, an average IoU of 0.89, and an F1-Score of 0.95. Compared to the method without the fractal dimension matrix but with image enhancement, the mean accuracy improved by 11.6 percentage points, the mean IoU increased by 0.28, and the F1-Score rose by 0.32. Using the fractal dimension matrix prompt without image enhancement resulted in a mean accuracy of 88.9 %. Although this was 2.9 percentage points higher than the method without the fractal dimension matrix, the mean IoU and F1-Score decreased by 0.06 and 0.07, respectively. This indicates that while the fractal dimension matrix prompt helps improve accuracy, image enhancement was crucial for maintaining high-quality detection results. Overall, the combination of the fractal dimension matrix prompt and image enhancement yields the best performance in crack segmentation tasks, significantly improving all evaluation indicators.

3.5. Comparative evaluation

To assess the effectiveness of the proposed method, this study compared it with mainstream supervised deep learning approaches. The computer configuration used in the experiment was Intel (R) Core (TM) i7–10,700 CPU @ 2.90 GHz, GPU was NVIDIA GeForce RTX 3060 (8 GB), and the operating system was Windows 11. And the settings of training hyperparameters: learning rate was 0.001, optimizer was adam (adaptive moment estimation), training epochs were 100, and minimum batch size was 8.

Table 3 presents the comparative results of the proposed method and several commonly used semantic segmentation models for land-based crack segmentation, the training sample for this supervised deep learning model of land cracks was 120. In order to prevent overfitting of

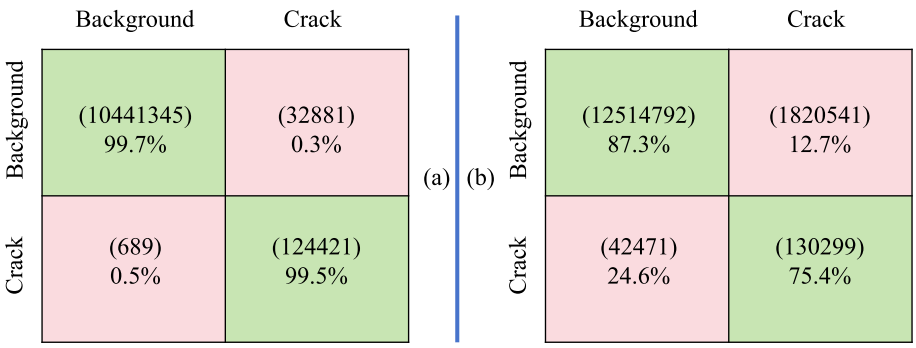


Fig. 14. Confusion matrix of detection accuracy.

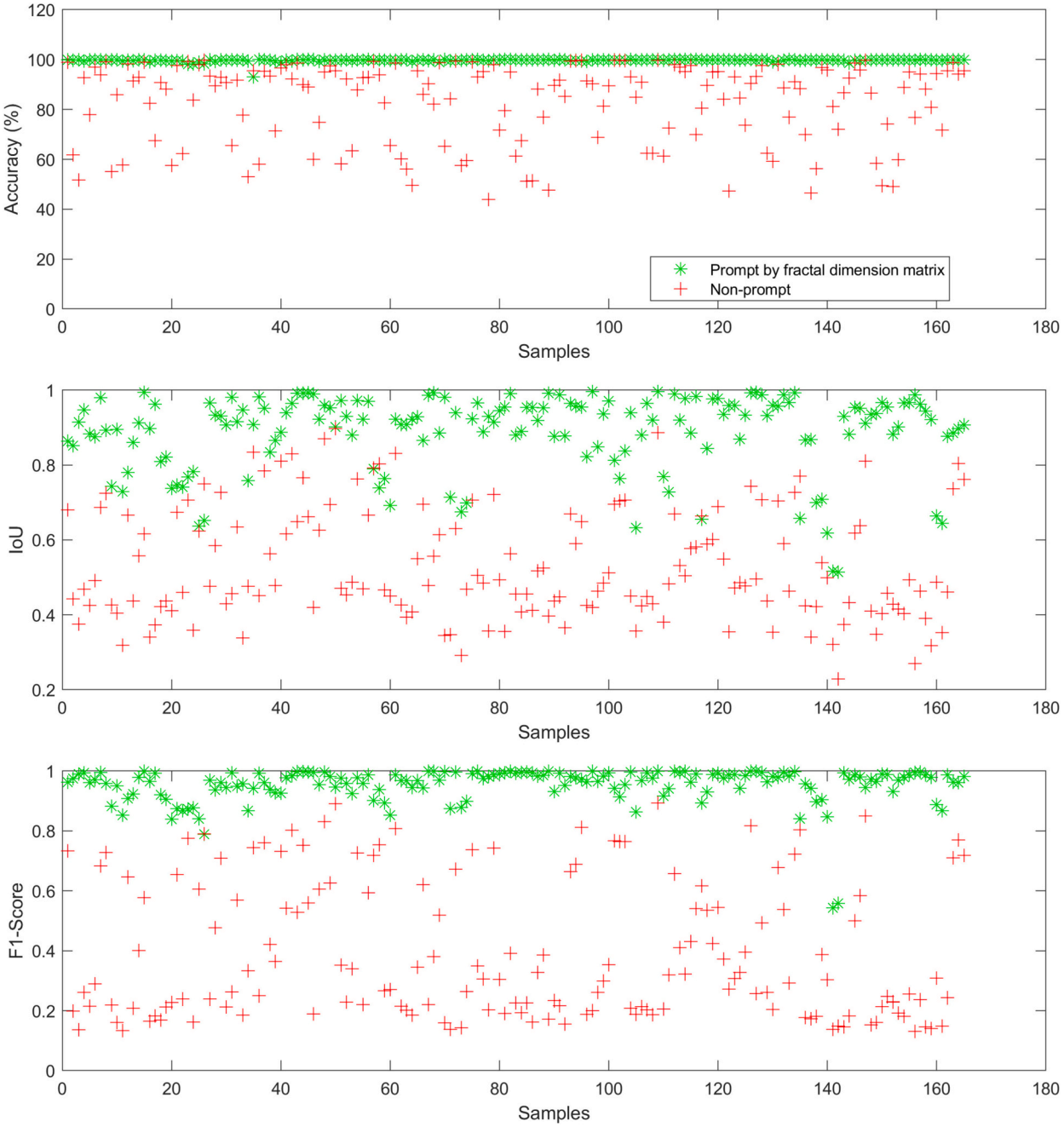


Fig. 15. Comparison of evaluation indicator values for all detected samples.

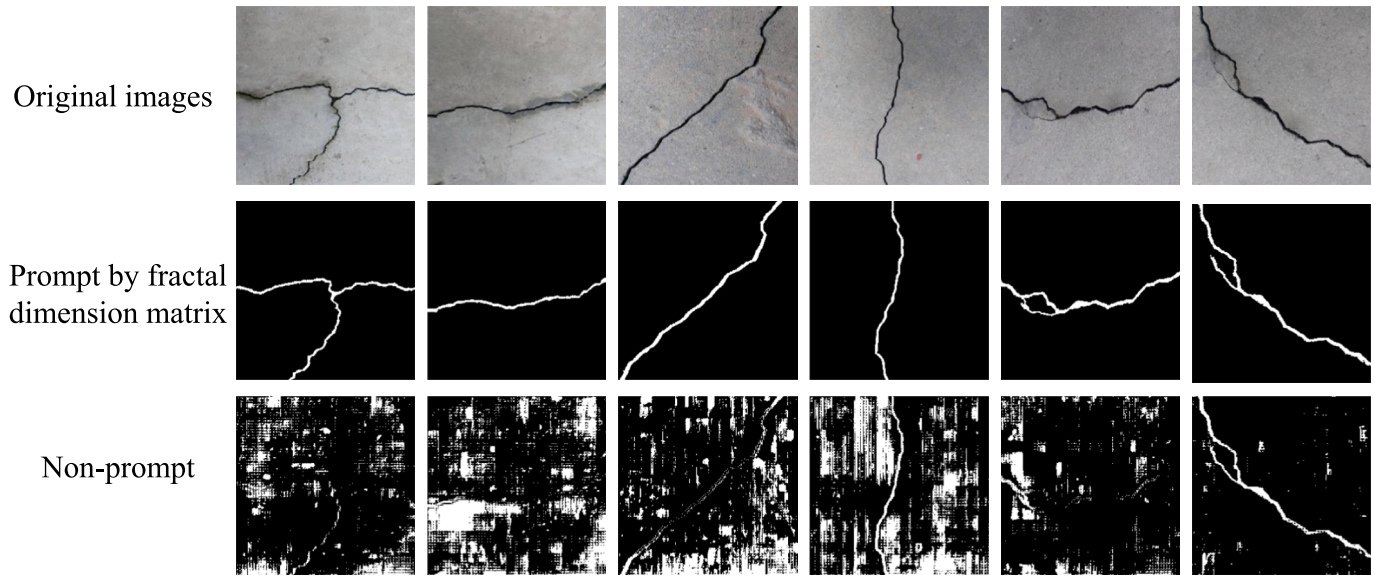


Fig. 16. Comparison case of detection effects.

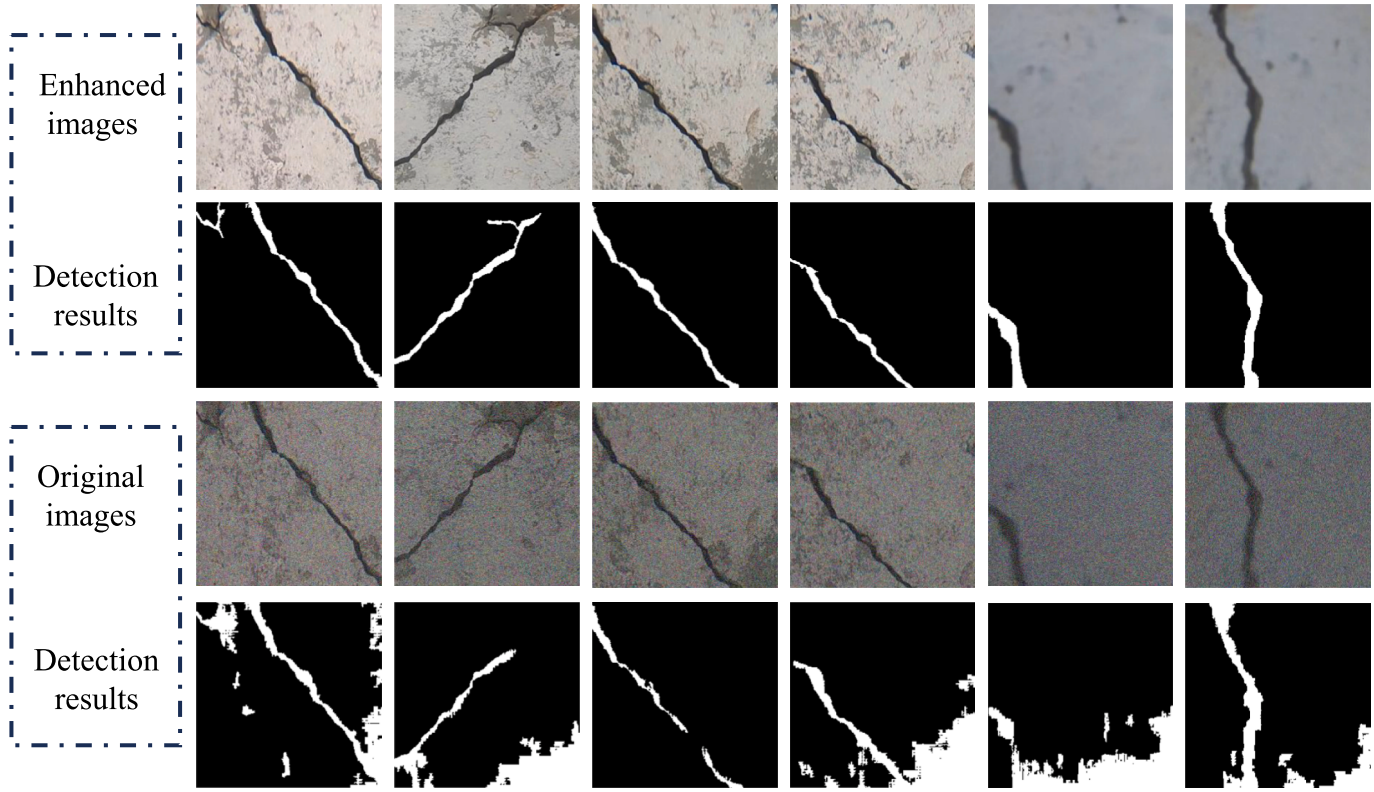


Fig. 17. Detection effect before and after underwater image enhancement.

the network model, random transformation functions (scaling, translation, and rotation) were added during the training process to enhance sample diversity. Meanwhile, in order to validate that the trained model did not overfit, K-fold cross validation was also conducted. In terms of average accuracy, the proposed method achieves the highest accuracy (99.6 %), followed by U-Net (97.2 %), Segnet and HRNetV2 (both 97.1 %). CAT-Seg has the lowest accuracy, only 75.3 %. Regarding average IoU, the proposed method, DeepLabv3+, Segnet, and U-Net all reach 0.89, exhibiting similar performance, while HRNetV2 was slightly lower at 0.88. CAT-Seg has the lowest IoU at 0.21. In F1-Score, the proposed method, DeepLabv3+, Segnet, and U-Net all achieve 0.95, with similar performance, while HRNetV2 was slightly lower at 0.94. CAT-Seg has the lowest F1-Score at 0.46. The proposed method demonstrates excellent performance in terms of average accuracy, average IoU, and F1-Score, comparable to or better than other mainstream methods such as DeepLabv3+, Segnet, U-Net, HRNetV2, OpenTrans, EBSeg and significantly outperforms FCN and CAT-Seg. Some detection cases were shown in Fig. 19, the results indicate that these compared models tend to overlook many details in crack segmentation, especially CAT-Seg's crack segmentation was too rough.

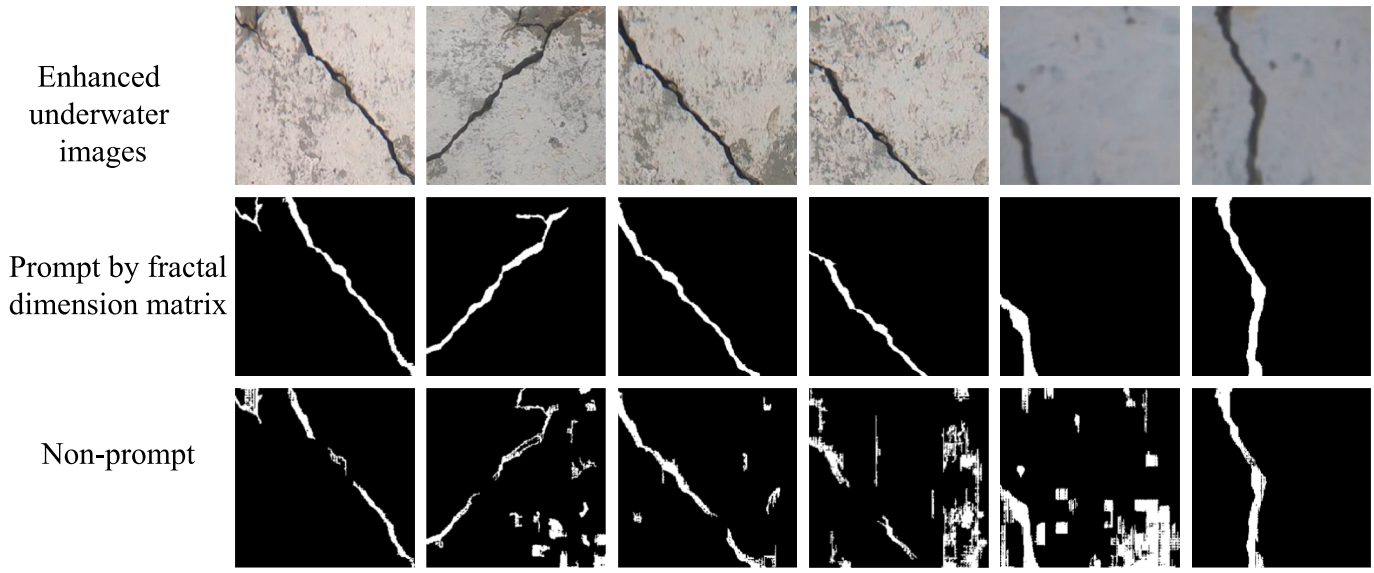


Fig. 18. Comparison case of detection effects.

Table 2
Comparison of underwater crack detection results.

	Evaluation indicator		
	Mean accuracy	Mean IoU	F1-Score
Prompt by fractal dimension matrix (image enhancement)	97.6 %	0.89	0.95
Prompt by fractal dimension matrix (Original image)	88.9 %	0.55	0.56
Non-prompt (image enhancement)	86.0 %	0.61	0.63

Table 3
Comparison of land crack detection results.

	Evaluation indicator				
	Mean accuracy	Mean IoU	F1-Score	Training times	Detection times
Proposed method	99.6 %	0.89	0.95	None	26 s
DeepLabv3+	95.2 %	0.89	0.95	247 s	16 s
Segnet	97.1 %	0.89	0.95	291 s	22 s
FCN	80.5 %	0.62	0.81	7085 s	85 s
U-Net	97.2 %	0.89	0.95	1308 s	85 s
HRNetV2	97.1 %	0.88	0.94	242 s	9 s
OpenTrans	95.3 %	0.86	0.88	376 s	25 s
EBSeg	91.7 %	0.80	0.89	790 s	52 s
CAT-Seg	75.3 %	0.46	0.21	638 s	36 s

In terms of time efficiency, the proposed method exhibits significant advantages. Unlike other supervised deep learning methods that require extensive training time for crack feature learning, the proposed method can perform crack segmentation tasks without network training. The total time required for detecting all the samples was only 26 s. In contrast, other deep learning methods, especially FCN, require much longer training and detection times due to the crack feature learning process. This time comparison confirms the exceptional detection efficiency of the proposed method while also avoiding the cost of data annotation. Since there are limited underwater crack images available for training, it's not feasible to compare the detection results of underwater crack images with those of other mainstream deep learning models. However, this also demonstrates the outstanding capability of the proposed method to address the challenge of insufficient sample data.

The proposed method for crack segmentation offers significant advantages over other supervised deep learning models that require training. Firstly, the SAM model was a pre-trained segmentation model, which was plug-and-play and does not require additional training for specific tasks, thus drastically reducing development and deployment time. Secondly, the SAM model demonstrates excellent generalization ability and can perform well across various scenarios without the need for extensive annotated data. Moreover, the fractal dimension matrix provides SAM with precise prompt information, enabling it to achieve high-precision crack detection results. Its versatility and efficient inference speed make it more flexible and effective in practical applications.

In order to further confirm the effectiveness of the plug-and-play model proposed in this paper, it was also compared with some other training-free models, Due to the fact that training-free models do not require training samples, underwater crack images with fewer samples were also compared in this comparison. Table 4 shows the comparative results of training-free model, indicating that the proposed method has strong detection performance, the proposed method has the highest detection accuracy in both land and underwater crack images, and its detection speed was not inferior to other models. These detection results further confirm the enormous advantages of the proposed method.

4. Conclusions

The SAM based on fractal dimension matrix prompts demonstrates significant advantages in crack segmentation tasks. Through the fractal dimension matrix prompt, this method effectively enhances detection accuracy and efficiency, compared to other traditional methods, our approach not only reduces development and deployment time but also improves the reliability and precision of detection results. Additionally, image enhancement techniques such as image inversion and haze removal significantly improve the quality of underwater crack images, making crack details clearer. The proposed method features plug-and-play characteristics, requiring no additional training for specific tasks, and possesses good generalization ability and efficient inference speed, making it more flexible and efficient in practical applications. Overall, our approach provides an efficient and accurate solution for the detection of both onshore and underwater concrete cracks, with significant practical value.

Based on the research findings, the following conclusions can be drawn:

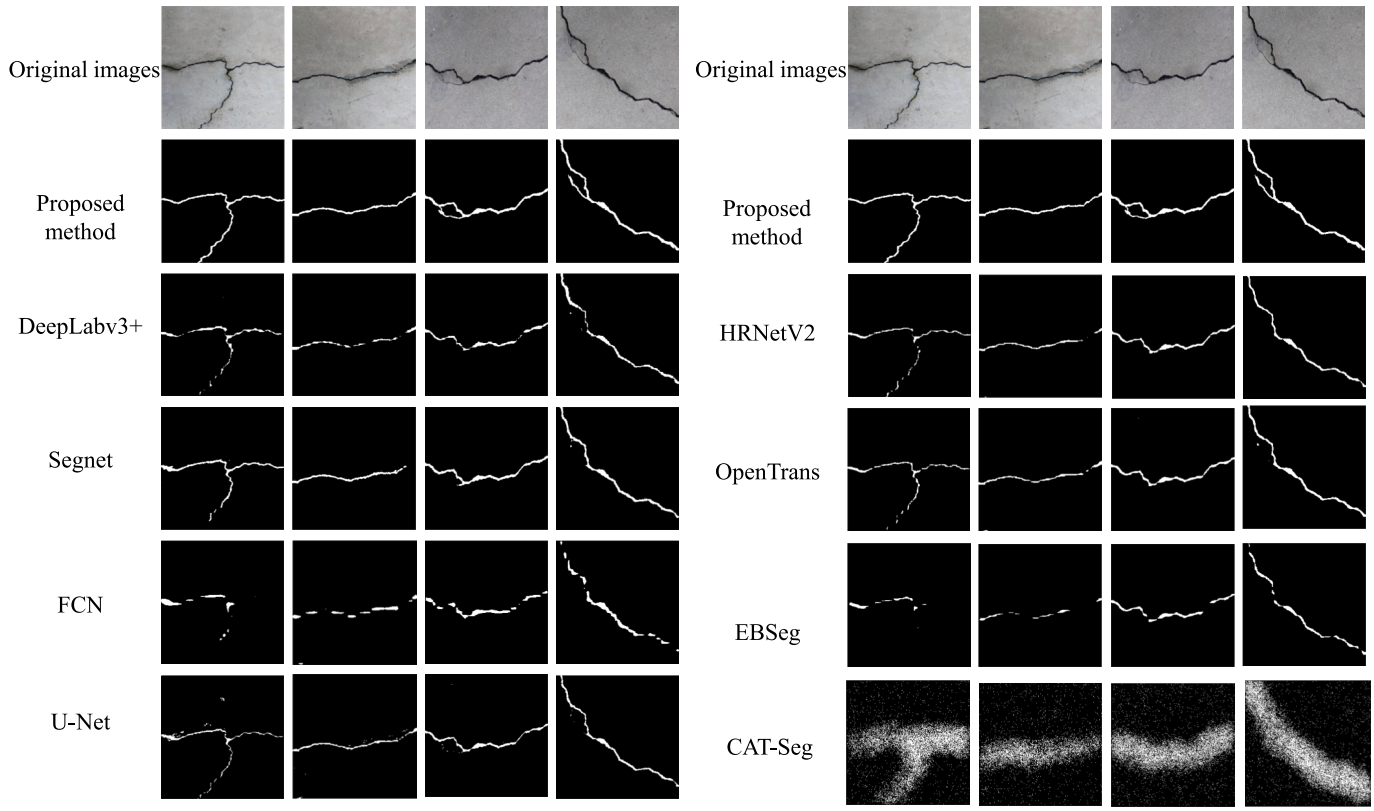


Fig. 19. Examples of segmentation performance of different network models.

Table 4
Comparative results of training-free model.

		Evaluation indicator				
		Mean accuracy	Mean IoU	F1-Score	Training times	Detection times
Land crack	Proposed method	99.6 %	0.89	0.95	None	26 s
	FreeDA	86.4 %	0.78	0.89	None	38 s
	MaskDiffusion	84.1 %	0.68	0.73	None	33 s
	DiffSegmenter	67.3 %	0.49	0.36	None	26 s
Underwater crack	Proposed method	97.6 %	0.89	0.95	None	1 s
	FreeDA	76.8 %	0.63	0.72	None	2 s
	MaskDiffusion	79.2 %	0.69	0.75	None	s s
	DiffSegmenter	70.2 %	0.59	0.66	None	1 s

(1) The SAM based on fractal dimension matrix prompts performs exceptionally well in crack segmentation tasks. Experimental results demonstrate that for onshore crack segmentation, the mean accuracy, mean IoU, and F1-Score reached 99.6 %, 0.89, and 0.95, respectively.
(2) The proposed method maintains good detection performance in underwater crack segmentation tasks. The mean accuracy, mean IoU, and F1-Score achieved were 97.6 %, 0.89, and 0.95, respectively.
(3) Image enhancement techniques such as image inversion and haze removal significantly improve the brightness and contrast of underwater crack images, making crack details clearer and facilitating further detection and analysis.
(4) The proposed method does not require additional training for specific tasks and exhibits good generalization ability and efficient inference speed, making it more flexible and efficient in practical applications.
(5) By calculating fractal dimensions at multiple scales and spatial regions, the fractal dimension matrix provides a more comprehensive, detailed, and robust analysis of complex structures, effectively enhancing the accuracy and reliability of crack segmentation.

Although the method proposed in this paper achieves high-precision crack detection results and was more in line with practical engineering application requirements in a plug-and-play manner, there were still some limitations: (1) the universality of the method has not been validated, and the currently proposed method was only used to detect cracks on land and underwater, without validating its feasibility for other defects in bridges such as peeling and exposed reinforcement; (2) Lack of real-time capability. Currently, this method only analyzes local images and cannot achieve real-time monitoring of images. In the future, various types of defects will be analyzed to determine a universal model for detecting various types of defects in bridges, and a real-time segmentation system will be formed through the integration of algorithms and hardware, further enhancing the practical engineering application value of the proposed method.

CRediT authorship contribution statement

Shuai Teng: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Airong Liu:** Supervision, Resources, Methodology, Investigation, Funding acquisition, Conceptualization. **Zuxiang Situ:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Bingcong Chen:** Writing – review & editing, Project administration, Investigation. **Zhihua Wu:** Writing – original draft, Investigation, Formal analysis. **Yixiao Zhang:** Writing – original draft, Investigation, Formal analysis. **Jialin Wang:** Writing – review & editing, Supervision, Resources, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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